

SCIENTIFIC ALGORITHMS SPECIFICATION

Trustworthy AI for Protein Engineering

Discovering Mutation-Space Reliability and Applying It to Industrial Carbon-Capture Enzyme Design

Version 2.0

Document Purpose

This document defines every scientific calculation used throughout the project.

The Final Project Blueprint explains:

Why the science exists.

The Master Implementation Specification explains:

What software must be built.

This document explains:

Exactly how every scientific quantity is calculated.

The objectives of this document are:

- Ensure scientific reproducibility
- Remove algorithmic ambiguity
- Define all computational procedures
- Standardize calculations across all experiments
- Create a single source of algorithmic truth

Every dataset, figure, analysis, and conclusion must trace back to algorithms defined in this document.

Scientific Calculation Hierarchy

The project follows the calculation flow:

Raw Data



Empirical Sparsity

Evolutionary Sparsity

Structural Sparsity



Combined Sparsity



Experiment 1

Antisymmetry Error



Experiment 2

Epistasis Error



Integrated Reliability Analysis



Reliability Score



Industrial Fitness Score



Candidate Ranking

SECTION 1

EMPIRICAL SPARSITY

Purpose

Empirical Sparsity measures how rarely a mutation is observed within large experimental datasets.

Conceptually:

Frequently Observed Mutation

↓

Low Sparsity

Rarely Observed Mutation

↓

High Sparsity

Inputs

For each mutation:

Mutation ID

Observed Frequency

Total Mutation Count

Frequency Calculation

For mutation i:

Frequency(i)

=

Occurrences(i)

/

Total Observations

Empirical Sparsity Calculation

Empirical Sparsity is defined as:

Empirical Sparsity

=

1 - Normalized Frequency

where:

0

=

Most Common Mutation

1

=

Rarest Mutation

Interpretation

0.0 → Dense Region

0.5 → Moderate Density

1.0 → Extremely Sparse Region

SECTION 2

EVOLUTIONARY SPARSITY

Purpose

Evolutionary Sparsity measures how unusual a mutation appears from an evolutionary perspective.

This quantity is derived from ESM2 probabilities.

Inputs

Protein Sequence

Mutation

ESM2 Model

ESM Probability

For each mutation:

$P(\text{mutant amino acid})$

is generated by ESM2.

Evolutionary Sparsity Calculation

First:

Log Probability

=

$-\log(P)$

Then values are normalized:

0

=

Common Mutation

1

=

Rare Mutation

Interpretation

Low Evolutionary Sparsity

↓

Evolutionarily Plausible
High Evolutionary Sparsity

↓

Evolutionarily Unusual

SECTION 3

STRUCTURAL SPARSITY

Purpose

Structural Sparsity measures how constrained the structural environment is around a residue.

Inputs

Protein Structure

Residue Position

SASA Measurement

SASA Calculation

For each residue:

SASA

=

Solvent Accessible Surface Area

is calculated.

Structural Sparsity

Structural Sparsity is defined using normalized burial.

Conceptually:

Buried Residue

↓

High Sparsity

Exposed Residue

↓

Low Sparsity

Formula:

Structural Sparsity

=

1 - Normalized SASA

Interpretation

0.0 → Highly Exposed

1.0 → Highly Buried

SECTION 4

COMBINED SPARSITY

Purpose

Combine multiple sparsity perspectives into a single scientifically interpretable metric.

Benchmark Datasets

For Megascale-D studies:

Combined Sparsity

=

(Empirical
+ Evolutionary
+ Structural)

/ 3

TKT Industrial Application

Because empirical frequency information is unavailable:

Combined Sparsity

=

(Evolutionary
+ Structural)

/ 2

Interpretation

0.00 – 0.25
Very Dense

0.25 – 0.50
Moderately Dense

0.50 – 0.75
Sparse

0.75 – 1.00
Highly Sparse

SECTION 5

ANTISYMMETRY ERROR

Purpose

Measure thermodynamic inconsistency.

Inputs

For each mutation:

Forward Prediction

WT → MUT

and:

Reverse Prediction

MUT → WT

Thermodynamic Expectation

Physically:

$\Delta\Delta G(A \rightarrow B)$

=

$-\Delta\Delta G(B \rightarrow A)$

,

Antisymmetry Error

Formula:

Antisymmetry Error

=

| Forward + Reverse |

Examples

Perfect:

Forward
= +1.5

Reverse
= -1.5

Error
= 0

Moderate:

Forward
= +1.8

Reverse
= -1.5

Error
= 0.3

Large Failure:

Forward
= +2.4

Reverse
= -0.4

Error
= 2.0

Interpretation

0.0
Excellent Consistency

0.1–0.5
Minor Inconsistency

0.5–1.0

Moderate Inconsistency

> 1.0

Major Inconsistency

SECTION 6

EPISTASIS

Purpose

Quantify mutation interactions.

Inputs

Single Mutation A

Single Mutation B

Double Mutation AB

Experimental Epistasis

Formula:

Epistasis

=

$\Delta\Delta G_{AB}$

-

$(\Delta\Delta GA + \Delta\Delta GB)$

Predicted Epistasis

Identical calculation using predicted values.

SECTION 7

EPISTASIS ERROR

Purpose

Measure prediction failure relative to experiment.

Formula

Epistasis Error

=

| Predicted Epistasis

-

Experimental Epistasis |

Interpretation

Low Error
High Accuracy
High Error
Low Accuracy

SECTION 8

INTEGRATED RELIABILITY ANALYSIS

Purpose

Determine whether sparsity predicts prediction failure through thermodynamic inconsistency.

Scientific Chain

Combined Sparsity

↓

Antisymmetry Error

↓

Epistasis Error

Required Relationships

Analysis 1:

Combined Sparsity

vs

Antisymmetry Error

Analysis 2:

Combined Sparsity

vs

Epistasis Error

Analysis 3:

Antisymmetry Error

vs

Epistasis Error

Expected Scientific Outcome

The ideal pattern is:

Higher Sparsity

↓

Higher Antisymmetry Error

↓

Higher Epistasis Error

SECTION 9

RELIABILITY SCORE

Purpose

Transform sparsity into an intuitive engineering metric.

Design Philosophy

The Reliability Score must remain:

Simple

Interpretable

Explainable

Defensible

Complex machine-learning scoring systems are intentionally avoided.

Reliability Score Formula

Reliability Score

=

1

-

Combined Sparsity

Range

0.0 → Lowest Reliability

1.0 → Highest Reliability

Categories

0.75 – 1.00

High Reliability
0.50 – 0.75

Moderate Reliability
0.25 – 0.50

Low Reliability
0.00 – 0.25

Very Low Reliability

Interpretation

The score answers:

How much confidence
should we place in
this prediction?

It does not answer:

Is the mutation beneficial?

SECTION 10

ACTIVE-SITE DISTANCE

Purpose

Control for structural importance.

Inputs

Protein Structure

Active-Site Residues

Mutation Position

Calculation

Minimum spatial distance between:

Mutation Residue

and:

Any Active-Site Residue

Interpretation

Small Distance

=

Near Active Site

Large Distance

=

Far From Active Site

SECTION 11

PREDICTED STABILITY SCORE

Purpose

Provide a normalized measure of predicted mutation benefit.

Inputs

FoldX

DDMut

RaSP

Other Predictors

Consensus Stability

If multiple predictors are available:

Consensus Stability

=

Average

Normalized Prediction

Interpretation

High Score

=

Predicted Stability Gain

Low Score

=

Predicted Stability Loss

SECTION 12

INDUSTRIAL FITNESS SCORE

Purpose

Rank mutations for future engineering investigation.

Philosophy

The project does not attempt to identify:

The Best Mutation

Instead it identifies:

The Most Promising
High-Confidence Mutation

Inputs

Predicted Stability Score

Reliability Score

Evolutionary Plausibility

Structural Context

Initial Version

To maintain simplicity:

Industrial Fitness Score

=

$0.50 \times \text{Stability}$

+

$0.35 \times \text{Reliability}$

+

$0.15 \times \text{Evolutionary Plausibility}$

All inputs normalized to:

0–1

Interpretation

High Industrial Fitness requires:

Good Predicted Stability

High Reliability

Reasonable Evolutionary Support

A mutation with extreme predicted benefit but very poor reliability should rank lower than a mutation with slightly lower benefit but much higher confidence.

SECTION 13

CANDIDATE RANKING

Purpose

Create a prioritized list of mutations.

Procedure

For every TKT mutation:

Calculate:

Combined Sparsity

Reliability Score

Predicted Stability

Industrial Fitness Score

Then:

Sort

Descending

By

Industrial Fitness Score

Final Output

Rank

Mutation

Reliability

Predicted Stability

Industrial Fitness Score

SECTION 14

THE KILLER FIGURE ALGORITHM

Purpose

Generate the flagship figure of the project.

Required Components

The figure must visually connect:

Megascale-D



Combined Sparsity



Antisymmetry Error



Epistasis Error



Reliability Framework



Complete TKT Landscape



Industrial Fitness Ranking



Top Candidate Mutations

End Of Scientific Algorithms Specification

This document defines the complete scientific logic of the project.

Every dataset, figure, analysis, and conclusion must be traceable to algorithms defined within this specification.

The next document to update is:

DATA ARCHITECTURE &
DATASET SPECIFICATION

which will formally define every dataset produced by these algorithms.